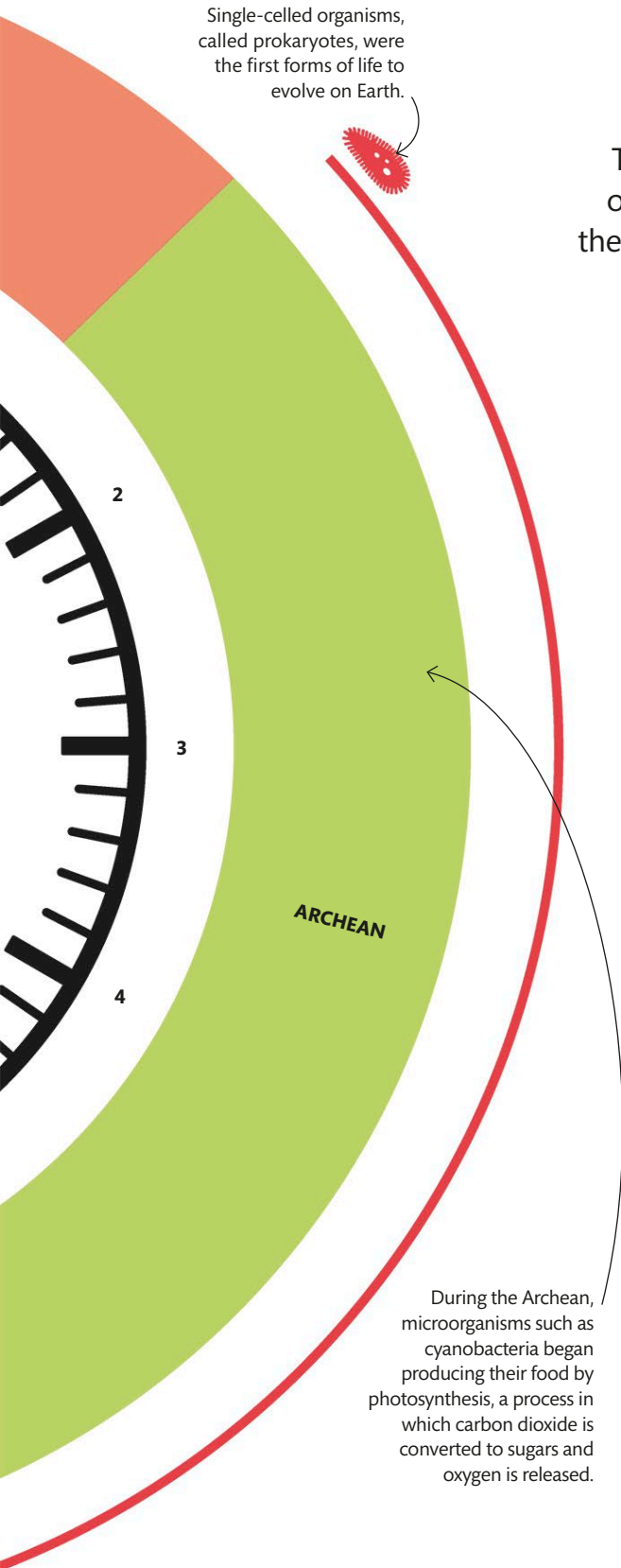


HISTORY OF LIFE ON EARTH

Single-celled organisms, called prokaryotes, were the first forms of life to evolve on Earth.



During the Archean, microorganisms such as cyanobacteria began producing their food by photosynthesis, a process in which carbon dioxide is converted to sugars and oxygen is released.

Planet Earth formed about 4.5 billion years ago. Although scientists still debate about when life first evolved on Earth, many think that the first life forms may have appeared shortly after the oceans formed around 4 billion years ago. These organisms were tiny single-celled beings called prokaryotes. More complex organisms began to evolve a few billion years later, and these in turn evolved into the animals, plants, and fungi we see today. Although 99 percent of all the life forms that have ever existed on Earth are now extinct, all known life forms—both fossil and living today—share an ancestry that can be traced back into deep time.

THE ERAS IN THIS BOOK

PALEOZOIC (BEFORE THE DINOSAURS)

This era saw the evolution of a wide range of creatures, including fish and amphibians, as well as the emergence of the first reptiles and the primitive relatives of modern mammals. It is the longest of the eras featured in this book and ended with a catastrophic mass extinction that destroyed 96 percent of all species.



Brontoscorpio

MESOZOIC (AGE OF THE DINOSAURS)

Dinosaurs first appeared during the Mesozoic, which is divided into three distinct periods.

Triassic Period

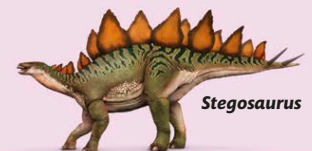
The first dinosaurs were small, agile creatures. They coexisted with many larger types of animals until an extinction event at the end of the period wiped out most other life forms, allowing the dinosaurs to dominate on land.



Herrerasaurus

Jurassic Period

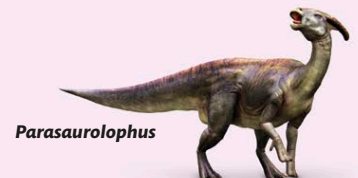
During the Jurassic, the dinosaurs ruled the land and evolved into many large and fearsome forms. This period also saw the appearance of the first birds and some early mammals.



Stegosaurus

Cretaceous Period

Dinosaurs flourished in this period, which saw the evolution of many diverse species and the appearance of flowering plants. At the end of this period, an asteroid impact wiped out most dinosaurs from the face of Earth, leading to the rise of mammals.



Parasaurolophus

CENOZOIC (AFTER THE DINOSAURS)

In this period, birds—the only dinosaurs to survive the end-Cretaceous extinction—dominated the skies, while mammals thrived on land. Early humans took their first steps.



Mammut borsoni